

AI for Project Management

Comprehensive 15-Topic Curriculum:
From Introduction to Ethical AI

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Understanding AI in PM



AI empowers project managers to automate decisions, predict outcomes, and optimize resources — transforming how projects are planned, executed, and delivered.

Understanding the Core Concepts

- ✱ **Artificial Intelligence (AI)**
The simulation of human intelligence by machines to perform tasks like reasoning, problem-solving, and decision-making in PM workflows.
- ✱ **Machine Learning (ML)**
A subset of AI where systems learn from project data to improve predictions — such as schedule delays, cost overruns, and resource needs.
- ✱ **Generative AI (GenAI)**
AI that generates new content — drafting charters, reports, risk logs, and meeting summaries — based on prompts from project managers.

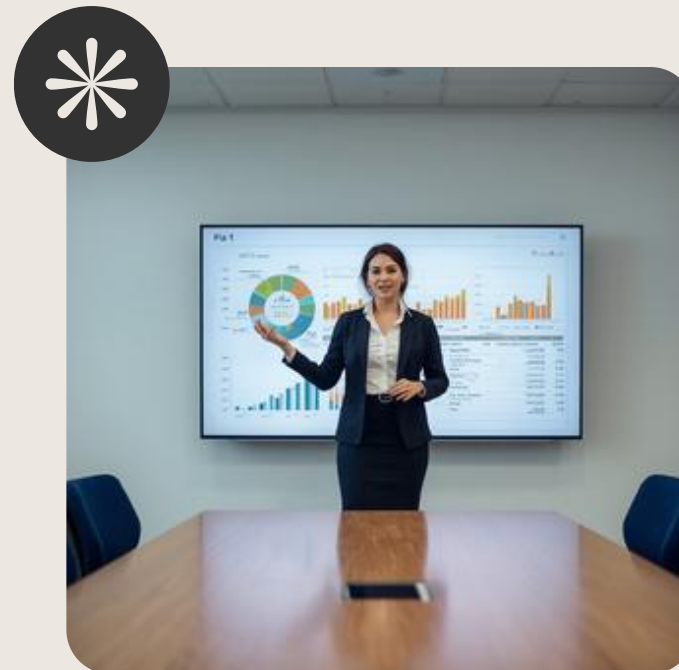
Current Landscape & Personas

Loren



Program Manager using AI for stakeholder mapping, automated reporting, and meeting transcription with Fathom.

Pia



Project Analyst leveraging AI for feasibility analysis, tone adjustment, and communication clarity with Notion AI.

Alan



Senior PM applying AI for charter development, quality test case generation, and scope validation workflows.



Prompt Engineering

Introduction & Anatomy

Crafting Inputs for Optimal AI Outputs: Prompt engineering is the practice of designing clear, structured instructions to guide AI models toward accurate and useful responses in project management tasks.

What is Prompt Engineering?

The skill of writing precise instructions for AI tools to generate relevant outputs — such as drafting charters, risk logs, or meeting summaries.

Anatomy of an Effective Prompt

A strong prompt includes: Role ("Act as a PM"), Context (project details), Task (what to generate), and Format (output structure).

Simple Prompt Examples

"List 5 risks for a software migration project." or "Draft a project charter for a CRM rollout with a 3-month timeline."

Iterative Prompting

STEPWISE REFINEMENT
FOR OPTIMAL AI
OUTPUTS



Step 1

Draft Initial Prompt



Step 2

Review AI Output & Identify Gaps



Step 3

Refine Prompt with Context &
Constraints



Step 4

Validate Output – Persona: Pia's Risk
Log



Hands-On Lab: Prompt Engineering



Persona: Sonny is a project manager who needs to generate and refine AI prompts for task planning. Follow each step below to build and improve your prompts for AI-driven project task generation.



Step 1: Write a Base Prompt

Draft an initial prompt asking an AI tool to generate a task list for a software deployment project. Keep it simple and direct, e.g., "List tasks for deploying a CRM system."



Step 2: Add Context & Constraints

Refine your prompt by adding role, timeline, and scope. Example: "As a PM, list 10 tasks for deploying a CRM in 3 months with a 5-person team, grouped by phase."



Step 3: Iterate and Evaluate

Compare the outputs from Steps 1 and 2. Identify gaps, ambiguities, or missing deliverables. Refine the prompt once more to address these issues and improve output quality.



Step 4: Expected Outcomes

Produce a polished, structured task list suitable for a real PM plan. Reflect on how each iteration improved the AI output. Share your best prompt with the group for discussion.

Charter Development *AI for Project Initiation*



AI transforms charter development by auto-generating scope, objectives, and stakeholder roles. Alan uses AI to draft technical charters; Risa leverages it for stakeholder alignment and risk summaries — cutting charter prep time significantly.

* Key Charter Components

AI drafts project objectives, scope boundaries, deliverables, constraints, and success criteria from minimal inputs.

* Persona Examples

Alan: AI-assisted technical scope definition.
Risa: Automated stakeholder mapping and risk identification in charters.

* Hands-On Lab

Lab: Use an AI tool to generate a project charter from a brief scenario. Validate outputs and refine with iterative prompting.



Feasibility Analysis

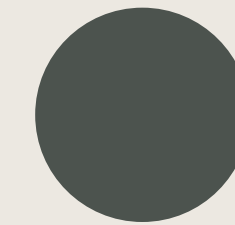
AI for Project Initiation

AI evaluates feasibility across multiple dimensions, helping project managers like Pia make faster, data-driven go/no-go decisions. Lab: Conduct an AI-assisted feasibility assessment.



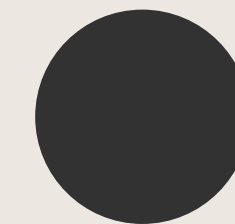
Key Factors AI Evaluates

Budget viability, resource availability, timeline constraints, technical complexity, and risk exposure analyzed instantly.



Persona Use Case: Pia's Scenario

Pia uses AI to assess a new digital transformation project, receiving a scored feasibility report in minutes.



Hands-On Lab Activity

Input project parameters into an AI tool, generate a feasibility report, and interpret the results for stakeholders.

AI for Project Initiation

STAKEHOLDER MAPPING WITH AI



Technique 1

AI-Powered Stakeholder Identification



Technique 2

Influence & Interest Matrix via AI Analysis



Technique 3

Persona Example: Loren Maps Project Stakeholders

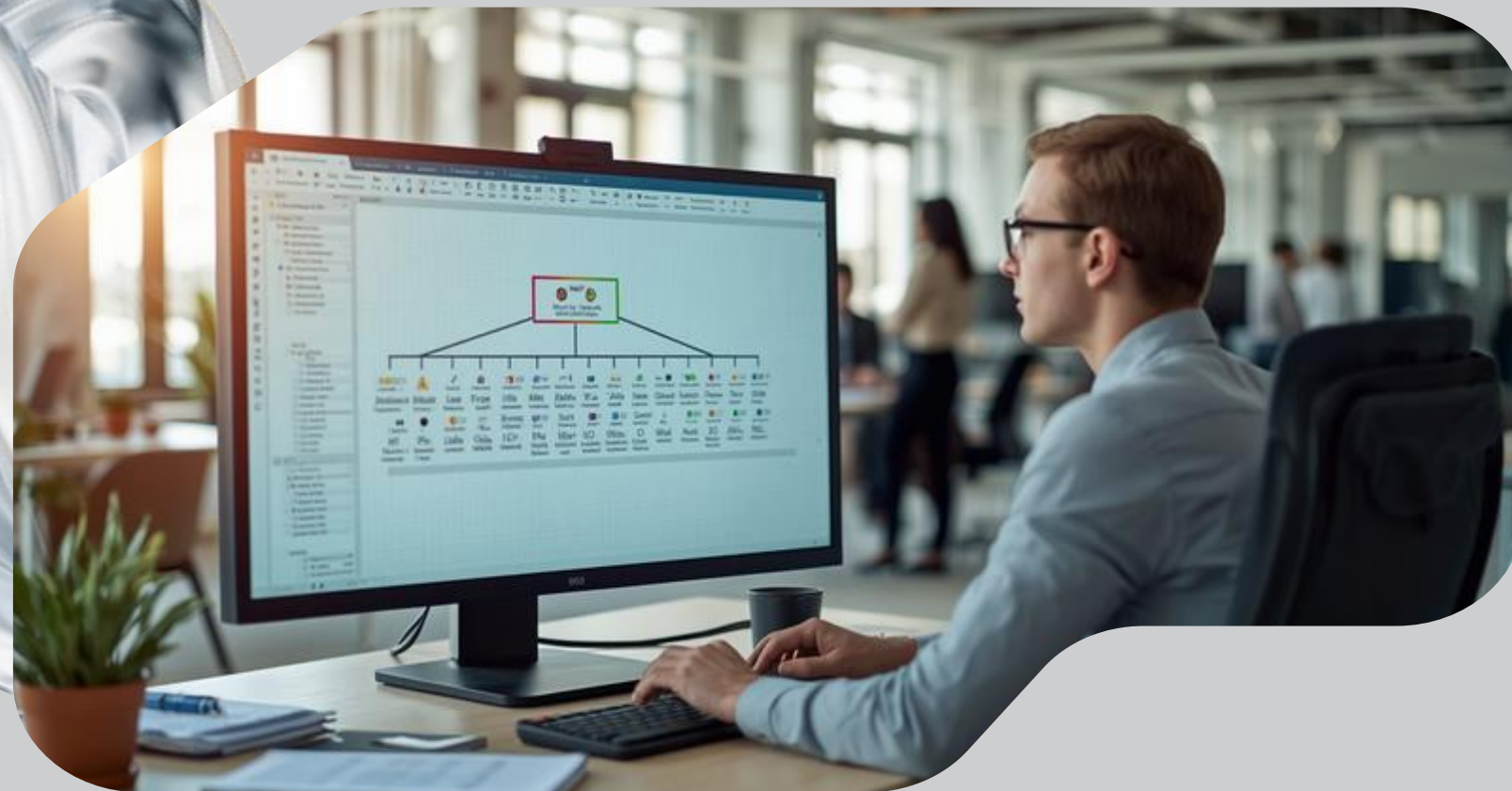


Lab Activity

Lab: Build a Stakeholder Map Using AI Tools



Work Breakdown Structure *Automating Scope Definition*



A well-structured WBS is the foundation of effective project planning. AI tools can instantly decompose project goals into detailed task hierarchies, identify missing deliverables, and validate structure — saving hours of manual effort.

* Why WBS Matters

WBS decomposes project deliverables into manageable tasks, enabling clear ownership, accurate estimation, and progress tracking.

* AI Techniques for WBS

AI generates hierarchical task structures from project descriptions and validates completeness against similar past projects.

* Hands-On Lab

Lab: Use AI to generate and validate a WBS for a sample project. Refine outputs using iterative prompting techniques.



Requirements Gathering

AI-Assisted Elicitation

AI transforms how teams elicit, document, and validate project requirements. Persona: Migz uses AI to streamline user story creation and close gaps in scope documentation.

AI Elicitation Methods

AI analyzes stakeholder inputs, past projects, and domain knowledge to surface complete, structured requirements automatically.

Natural Language Processing

NLP tools parse meeting transcripts and emails to extract and categorize functional and non-functional requirements.

Lab: User Story Generation

Migz uses an AI prompt to generate structured user stories from raw stakeholder notes, then refines them for sprint planning.

Scope Creep Detection

AI IDENTIFIES AND FLAGS
UNAUTHORIZED SCOPE CHANGES
BEFORE THEY DERAIL YOUR
PROJECT:



Definition: Scope creep is the uncontrolled expansion of project scope without adjustments to time, cost, or resources



AI Approach: Continuously compares deliverables against baseline scope, flagging deviations in real time



Before/After: Manual reviews missed 40% of creep; AI detection reduced scope overruns by 65%

Lab Activity: Using AI tools, analyze a sample project log to identify instances of scope creep. Compare baseline requirements against current deliverables, classify each deviation, and propose corrective actions to realign the project scope.



Schedule Optimization *Dynamic Timelines*



Dynamic scheduling allows AI to continuously adjust project timelines based on real-time data, resource availability, and shifting priorities — helping teams like Grace's stay on track even as conditions change.

* Dynamic Scheduling

AI monitors task progress and automatically reschedules milestones to reflect real-time project status and dependencies.

* Timeline Flexibility

AI adapts timelines when risks or delays arise, offering alternative paths while keeping delivery goals intact.

* Lab: Gantt Chart Generation

Persona: Grace uses AI to generate and optimize a Gantt chart for her team's sprint cycle, balancing workload and deadlines.

Critical Path Method

AI-ENHANCED CRITICAL PATH OPTIMIZATION

🕒 Components

Tasks, dependencies, durations, float time & milestone sequencing

🕒 AI Advantages

AI auto-detects bottlenecks, recalculates float & flags delays in real time

🕒 Optimization

Dynamic re-routing of critical path when resources or timelines shift

🕒 Lab Activity

Lab: Critical path calculation practice using AI scheduling tools



Schedule Optimization: Resource Leveling

Resource leveling adjusts task schedules to resolve over-allocation and balance workloads. AI automates this process by analyzing capacity, dependencies, and priorities in real time.



What is Resource Leveling?

A scheduling technique that resolves resource conflicts by shifting tasks within float, ensuring no team member is over-allocated or underutilized.



Lab: Identify Overloads

Using a sample project dataset, prompt an AI tool to identify over-allocated resources and flag tasks eligible for rescheduling.



AI Advantages

AI scans all tasks, deadlines, and team capacity simultaneously — recommending optimal reassignments faster than manual methods.



Lab: Balance Resources

Apply AI-generated recommendations to rebalance workloads. Compare the before and after resource histogram to evaluate the outcome.

Skill Mapping

Resource Allocation & Capacity Planning



AI-powered skill mapping enables project managers to allocate the right people to the right tasks. By profiling team competencies, AI reduces mismatches, accelerates onboarding, and improves overall project performance.

- * **AI Skill Profiling**
AI analyzes team member profiles, past performance, and certifications to match skills to project tasks with precision.
- * **Persona: Win**
Win uses AI to map her team's competencies, identify skill gaps, and optimize task assignments across concurrent projects.
- * **Hands-On Lab**
Lab: Build a skill inventory using AI, identify gaps, and generate a recommended allocation plan for a sample project team.

Bottleneck Prediction

AI-POWERED EARLY DETECTION TO
PREVENT RESOURCE BOTTLENECKS
BEFORE THEY IMPACT DELIVERY:



AI scans workload data to
flag overloaded resources
and predict bottlenecks
weeks in advance



Persona – Ronaldo: Uses AI
dashboards to monitor team
capacity and redistribute tasks
proactively



Lab: Analyze a project
scenario and apply AI
to identify and resolve
a bottleneck

Early bottleneck identification is critical to on-time delivery. AI tools analyze task queues, resource utilization rates, and historical velocity data to surface risks before they escalate into project delays.



Burnout Prevention with Wrike

AI-POWERED WORKLOAD
BALANCE & TEAM
WELLBEING

Features Overview



Wrike AI monitors workloads in real time, flags overallocation, auto-redistributes tasks, and sends burnout risk alerts to managers.

Use Case Example



Persona – Risa: Uses Wrike's capacity heatmap to rebalance sprint tasks across his team, reducing overtime by 30% in two weeks.

Hands-On Lab



Lab: Simulate a team workload scenario in Wrike. Identify overloaded members, apply AI suggestions, and document the rebalancing outcome.

AI in Budgeting



Sonny uses AI tools to draft accurate budget baselines by analyzing historical project data, estimating costs across phases, and flagging potential overruns early — enabling proactive financial control.

Baseline Drafting

* Budget Baseline Components

Cost estimates, resource rates, contingency reserves, and phased spending plans form the core of an AI-generated baseline.

* AI Methods for Baselining

AI analyzes historical data, benchmarks similar projects, and applies parametric modeling to produce accurate cost baselines.

* Lab: Budget Baselining

Use AI prompts to generate a budget baseline for a sample project. Identify cost categories, assign estimates, and validate with AI feedback.

Hidden Cost Identification

AI TECHNIQUES TO UNCOVER
OVERLOOKED COSTS

🕒 Technique 1

Historical Data Mining

🕒 Technique 2

Anomaly Detection in Budget Patterns

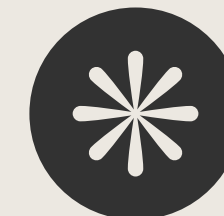
🕒 Technique 3

Vendor & Contract Cost Analysis



Lab

Cost Scenario Analysis Exercise



Scenario Planning with Idea Link

USING AI-POWERED
BUDGETING SCENARIOS TO
PLAN SMARTER AND ADAPT
FASTER.

Idea Link Overview



Idea Link enables teams to model multiple budget scenarios simultaneously, comparing outcomes and trade-offs in real time using AI-driven analysis.

Persona: Imee



Imee uses Idea Link to run best-case, worst-case, and most-likely budget scenarios for her infrastructure project, adjusting assumptions dynamically.

Hands-On Lab



Lab: Open Idea Link and create three budget scenarios for a sample project. Adjust cost variables and document the AI-generated insights for each scenario.

Risk Identification



Persona – Migz: As a risk manager, Migz uses AI to run structured brainstorming sessions, ensuring no risk category is overlooked due to cognitive bias or groupthink. AI surfaces hidden threats objectively.

🔬 Lab: Conduct an AI-facilitated risk brainstorming session for a sample project.

Unbiased Brainstorming

* Eliminates Cognitive Bias

AI evaluates risks neutrally, removing anchoring bias and groupthink from the brainstorming process.

* Broader Risk Coverage

AI cross-references historical data and industry patterns to surface risks humans may overlook.

* Structured Risk Taxonomy

AI categorizes risks by type, likelihood, and impact — enabling faster prioritization and response planning.

Automated Risk Scoring with Taskade

Taskade uses AI to automatically evaluate and score project risks. Follow these lab instructions to explore automated risk scoring and review expected outputs:



How Taskade Scores Risks

Taskade AI analyzes risk entries by evaluating likelihood, impact, and proximity. Each risk receives a numeric score and priority label (Low, Medium, High, Critical) automatically.



Run Automated Scoring

Prompt Taskade AI to score all listed risks. Review the generated scores, priority rankings, and suggested mitigation actions for each risk item in the register.



Lab Setup Instructions

Open Taskade and create a new Risk Register project. Input at least 5 project risks with brief descriptions. Enable the AI Risk Scoring feature from the project settings panel.



Expected Lab Outputs

A completed risk register with AI-assigned scores, a prioritized risk matrix, and at least 3 AI-generated mitigation recommendations ready for stakeholder review.





Daily Task Automation

Zapier Triggers & Actions

Grace uses Zapier to automate repetitive PM tasks, freeing time for strategic work. Key concepts: triggers, actions, and multi-step workflows.

Common Triggers in PM

New task created, status change, due date approaching, form submission, or email received in project inbox.

Automated Actions

Notify team via Slack, update spreadsheet, assign tasks, log time entries, or send stakeholder status emails automatically.

Lab: Build a Zapier Automation

Persona Grace: Create a multi-step Zap linking a project tool trigger to Slack and Google Sheets actions.

Zapier Hands-On Lab

Build real automation workflows using Zapier. Follow the step-by-step instructions below to create and test your first AI-powered task automation:



Step 1: Set Up Your Zap

Log in to Zapier and click "Create Zap." Choose a Trigger App (e.g., Gmail or Google Forms) and select a trigger event such as "New Email" or "New Form Response."



Step 3: Test & Activate

Run a test to confirm data flows correctly between apps. Review the output, fix any field mismatches, then turn on your Zap to activate live automation.



Step 2: Configure the Action

Select an Action App (e.g., Trello, Asana, or Slack). Map the trigger data fields to the action fields — for example, create a new task card from a form submission.



Step 4: Build a PM Workflow

Challenge: Automate a full PM task — trigger on a new project request email, create a task in your PM tool, and notify the team in Slack. Document your workflow.

AI for Meeting Management

FATHOM TRANSCRIPTION:
CAPTURING EVERY WORD,
AUTOMATICALLY

Key Benefits



Auto-transcription, real-time summaries, searchable meeting history, and seamless CRM integration for zero missed details.

Persona: Loren



Loren uses Fathom to transcribe client calls, extract action items instantly, and share summaries with her project team.

Lab Activity



Lab: Connect Fathom to your calendar, record a mock meeting, and review the AI-generated transcript and action items.

Action Items & Agendas

AI-ASSISTED MEETING
WORKFLOW

AI Agenda Generation Overview

Workflow: Transcript to Action Items

Lab: Generate Agenda & Action List

Review & Discussion



Tone Adjustment



Persona – Pia: As a project manager, Pia uses AI to tailor her written communication for different audiences, ensuring messages are clear, professional, and appropriately toned for executives, team members, and clients.

AI Communication Tools

* What Is Tone Adjustment?

AI tools analyze and rewrite messages to match the desired tone—formal, empathetic, assertive, or concise—based on audience and context.

* Key Use Cases

Softening escalation emails, formalizing status updates for executives, and simplifying technical language for non-technical stakeholders.

* Hands-On Lab

Lab: Take three sample project messages and use an AI tool to rewrite each in a different tone. Compare outputs and evaluate effectiveness.

Jargon Translation with Notion AI

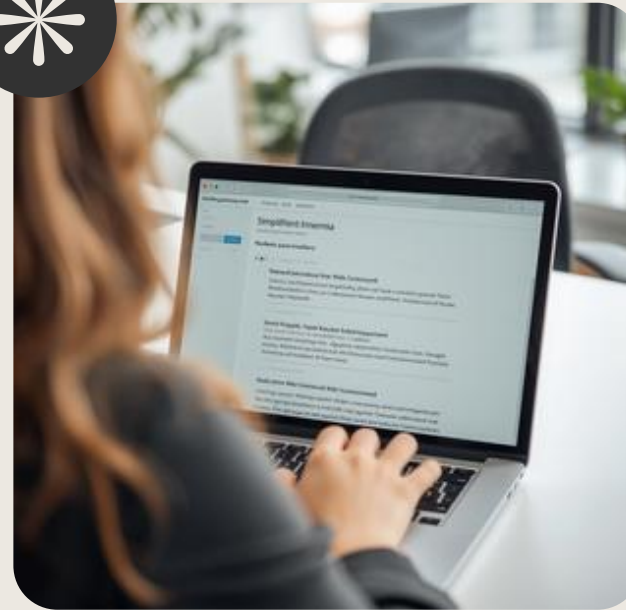
SIMPLIFYING TECHNICAL
LANGUAGE SO EVERY
STAKEHOLDER UNDERSTANDS:

The Challenge



Technical PM jargon creates barriers. Stakeholders misinterpret reports, leading to misaligned expectations and poor decisions.

Notion AI in Action



Notion AI rewrites complex project language into clear, audience-appropriate messaging. Persona: Pia uses it to translate sprint reports for executives.

Hands-On Lab



Build a Communication Hub in Notion AI. Translate a technical status report into plain language for three different stakeholder audiences.



Test Case Generation

AI-Powered QA Methods

Persona: Alan uses AI tools to generate comprehensive test cases, reducing manual effort and improving coverage. Lab: Generate test cases for a sample project module using AI.

AI Test Case Generation Methods

Boundary value analysis, equivalence partitioning, and exploratory test generation using AI prompts.

Defect Density Formula

Defect Density = Defects Found / Size of Module. AI tracks and benchmarks defect density across sprints.

Lab: Generate Test Cases

Use AI to auto-generate test cases from user stories. Review, refine, and validate outputs with Alan's workflow.

Documentation Review

AI-Powered Quality Assurance



Lab: Documentation Quality Check – Upload a sample project document and use an AI tool to review it for completeness, clarity, and compliance. Identify flagged issues and apply suggested corrections to improve overall documentation quality.

- ✱ **Error Detection**
AI scans project documents for inconsistencies, gaps, and compliance issues with speed and precision.
- ✱ **Standardization**
Ensures documentation follows organizational templates, terminology, and formatting standards automatically.
- ✱ **Continuous Improvement**
AI learns from past reviews to flag recurring issues and suggest best practices for future documentation.



Sentiment Analysis

Predictive Analytics

AI tools can detect emotional tone in team messages, feedback, and stakeholder communications — enabling proactive project leadership. Persona: Risa uses sentiment analysis to gauge team morale.

Understanding Sentiment Signals

AI scans text from emails, chats, and surveys to classify tone as positive, neutral, or negative in real time.

Stakeholder & Team Monitoring

Risa uses AI dashboards to track sentiment trends across stakeholders, flagging disengagement or concerns early.

Lab: Run a Sentiment Analysis

Input sample project communications into an AI sentiment tool and interpret the results to recommend PM actions.

Predictive Analytics – EVM

EARNED VALUE MANAGEMENT:
FORMULAS, AI INTEGRATION &
DELAY FORECASTING



Core EVM Formulas

$$SV = EV - PV$$

$$CV = EV - AC$$

$$SPI = EV / PV$$

$$CPI = EV / AC$$

$$EAC = BAC / CPI$$



AI Integration in EVM

AI auto-calculates EVM metrics in real time, flags negative SPI/CPI trends, and predicts project completion dates using historical velocity data



Lab: Delay Forecasting
Apply EVM formulas to a sample project dataset, use AI to forecast schedule delays and recommend corrective actions

EVM enables objective project performance measurement by comparing planned vs. actual vs. earned value. AI tools like ClickUp and Wrike automate EVM tracking, providing real-time variance alerts and predictive delay forecasting to keep projects on track.

ClickUp Brain

AI-POWERED FORECASTING AND
INTELLIGENCE FOR SMARTER PROJECT
DECISIONS

Key Capabilities



AI summaries, task predictions, auto-updates, deadline alerts, and intelligent project status reports — all in one platform.

Persona: Win



Win uses ClickUp Brain to forecast project completion dates, identify at-risk tasks, and auto-generate weekly status updates for stakeholders.

Hands-On Lab



Lab: Use ClickUp Brain to forecast a project timeline, generate a status report, and identify risks using AI-powered insights.

AI-Assisted Reporting

Status Compilation



Automated status compilation reduces reporting time by up to 70%. AI tools analyze live project data, highlight variances, and format reports tailored to stakeholder needs — ensuring accuracy and consistency across all reporting cycles.

* Automated Status Reports

AI aggregates project data, KPIs, and milestone updates into structured status reports — eliminating manual compilation effort.

* Persona: Bato's Workflow

Bato uses AI to pull task completion rates, budget variances, and risk flags automatically, generating weekly stakeholder reports in minutes.

* Lab: Compile Status Report

Use AI to compile a full project status report: input project data, select audience, and generate a formatted summary with key metrics and next steps.

Audience Tailoring with Gamma App

TAILOR STAKEHOLDER REPORTS
FOR DIFFERENT AUDIENCES USING
AI-POWERED FORMATTING AND
NARRATIVE TOOLS.

Audience Analysis



Gamma App identifies audience type — executive, technical, or operational — and adjusts tone, depth, and format accordingly.

Report Customization



AI restructures report content: executive summaries, visual dashboards, or detailed breakdowns — all from the same data source.

Lab: Executive Report



Lab: Take a standard project status report and use Gamma App to transform it into an executive-ready presentation format.



Enterprise vs Public AI

Ethical AI & Data Privacy

Sonny evaluates AI deployment options for his organization. Understanding the differences between enterprise and public AI is critical for responsible, secure project management.

Enterprise AI

Secure, private, customizable. Data stays within organizational boundaries with compliance controls and role-based access.

Public AI

Accessible and cost-effective, but carries risks: data exposure, lack of governance, and unpredictable outputs.

Choosing the Right Deployment

Evaluate sensitivity of data, compliance requirements, and team needs before selecting an AI deployment model.

Data Sanitization

TECHNIQUES TO PROTECT
DATA PRIVACY

🕒 Technique 01

Data Masking & Anonymization

🕒 Technique 02

Tokenization & Pseudonymization of Sensitive Fields

🕒 Technique 03

Redaction & Encryption Before AI Input

🕒 Lab Activity

Lab: Data Sanitization Exercise with AI Tools



Bias Mitigation

*DuckDuckGo
AI Chat*



Lab: Bias Auditing Practice — Use DuckDuckGo AI Chat to audit a sample project communication for bias. Identify biased language, suggest neutral alternatives, and document findings in a bias mitigation report.

* Bias Auditing Methods

Systematic review of AI outputs for language bias, demographic skew, and unfair assumptions using structured audit checklists.

* DuckDuckGo AI Tools

Privacy-focused AI chat for anonymous bias testing — submit project content without exposing sensitive organizational data.

* Mitigation Strategies

Reframe biased prompts, apply inclusive language guidelines, and validate AI recommendations with diverse stakeholder input.



Thank You & Questions

Thank you for your attention and participation throughout this curriculum.
For questions, next steps, or follow-up: connect with your facilitator or reach out via your course platform.

